

M2RIV Blueprint v3.0

The vendor-neutral release-evidence layer for deployable AI models

模型 release 應有一份標準化、可驗證、可攜、可被 CI 消費的 change evidence 。
那份 evidence 是 MCR ; M2RIV 是 reference implementation 。

要佔領的 abstraction

MCR 綁定 artifact identity、runtime provenance、paired evidence、uncertainty、policy、release authorization 與 build localization。成功不是所有人都執行 m2riv compare，而是其他 producer/consumer 不 import M2RIV Python 也能交換相同語意。

Producer / oracle	MCR boundary	Consumer
ModelOpt / Polygraphy	exact artifact + retained comparison + policy	CI / promotion
compiler CI / evaluator	content identity + PASS/WARN/BLOCK/ERROR	MLflow / registry
M2RIV reference CLI	conformance + strict verification	KServe / internal control plane

明確非目標

不做 evaluator zoo、training framework、registry、serving platform、dashboard 或 scheduler。Behavioral eval 是 external evidence source；Polygraphy 保持 backend oracle；MLflow 保持 experiment/lifecycle platform。

已完成的 kernel

29	4+4	629	4
public schemas	positive/negative profiles	GPU holdout cases	corpus entries

Layer	What exists now	Trust boundary
Evidence kernel	snapshots, manifests, stable/run identities, plans	content integrity ≠ authorship
MCR 0.4	evidence/report/run IDs + target root	self-consistency ≠ authorship
Conformance	PASS/WARN/BLOCK/ERROR + four negative vectors	interoperability ≠ model safety
Cross-language	Python/Node/Rust vectors + Python<->Rust MCR	repository-owned evidence
Integrations	Polygraphy producer + MLflow consumer	reference, not vendor endorsement
Security	bounded parsing, HMAC cache, traversal/secret defenses	isolated CI still required

Live target milestone

RTX 4060 Laptop GPU / driver 555.97 / TensorRT 10.4.0 / Polygraphy 0.53.4 : 四個 build 各 629/629 backend matches ; quality gate 為 PASS, PASS, BLOCK, BLOCK ; first bad = build-02 。此里程碑完成 repository-owned vertical , 但不算 independent reproduction 。

六個月只押三件事

Priority	Deliverable	Exit signal
1 · MCR	language-neutral spec, compatibility, conformance, governance	independent producer + consumer
2 · NVIDIA/compiler	exact GPU bundles, tactics/runtime/precision corpus	one independently rerun target bundle
3 · Adoption	LLM inference, CV/edge, compiler/runtime design partners	three retained real CI gates

開宗立派指標

第一個非作者維護的 tool 輸出 MCR ；第二個系統只 consume MCR 而不 import Python ；相鄰工具 issue/PR 討論輸出 MCR ；陌生作者維護 m2riv-* integration 。 這些都比 stars 更接近 abstraction ownership 。

Month 0 → Month 6

Metric	Now	M3	M6
External MCR producers	0	1	3
External MCR consumers	0	1	3
Active orgs with retained gates	0	2	5
Independently reproduced corpus cases	0	3	10
Unknown-to-founder integration maintainers	0	0	1

Repository-owned integrations and first-party GPU runs are tracked, but never counted as external adoption.

定位與決策護欄

Question	Start with	M2RIV relationship
Prompt / RAG / agent quality	Promptfoo / DeepEval / Inspect / Braintrust	consume their retained evidence
Experiment baseline validation	MLflow	log or promote verified MCR
TensorRT vs ORT output debug	Polygraphy	preserve raw result, add release semantics
Quantized artifact creation	ModelOpt	gate the produced build
Cross-tool release handoff	MCR	reference producer/verifier/conformance

Falsification

若 design partners 明確偏好 vendor-native reports、拒絕 portable promotion semantics、或相鄰標準形成廣泛採用的等價 contract、MCR moat hypothesis 便被削弱。應公開負訊號、不用擴 scope 掩蓋。

品牌 gate

M2RIV 仍是 pre-alpha working name。公開 v0.1 前完成 10–20 人 unaided spoken recall/spelling test、近似名稱與商標檢查。RFC 0015 已把 wire namespace 解耦為 mcr:sha256；品牌不應拖延驗證、但也不能由 CI 自動清除。

Canonical pitch

M2RIV turns model builds into reviewable release evidence.
Polygraphy tells you whether outputs differ. MCR tells the organization whether this exact build is releasable, why, with what evidence, and where the regression began.